

EE 115 Homework 8 Solutions

1. Filter output power

For a tone $A \cos(2\pi ft)$ or $A \sin(2\pi ft)$, $P = \frac{A^2}{2}$. The filter scales each tone by $|H(f)|$; phases do not change the power.

(a) $x(t) = 2 \cos(2\pi f_1 t) + 3 \sin(2\pi f_2 t)$

$$P_x = 2^2/2 + 3^2/2 = 6.5, \quad 8.13 \text{ dB.}$$

$$P_y = (2|H(f_1)|)^2/2 + (3|H(f_2)|)^2/2 = 0.2, \quad -6.99 \text{ dB.}$$

(b) $x(t) = 2 \cos(2\pi f_2 t) + 3 \sin(2\pi f_3 t)$

$$P_x = 6.5, \quad 8.13 \text{ dB.}$$

$$P_y = (2|H(f_2)|)^2/2 + (3|H(f_3)|)^2/2 = 0.08045, \quad -10.94 \text{ dB.}$$

(c) $x(t) = 3 \cos(2\pi f_1 t) + 4 \sin(2\pi f_2 t) + 5 \cos(2\pi f_3 t + \pi/3)$

$$P_x = 3^2/2 + 4^2/2 + 5^2/2 = 25, \quad 13.98 \text{ dB.}$$

$$P_y = (3|H(f_1)|)^2/2 + (4|H(f_2)|)^2/2 + (5|H(f_3)|)^2/2 = 0.36625, \quad -4.36 \text{ dB.}$$

2. Superhet receiver

Given $f_{\text{IF}} = 445 \text{ kHz}$, desired $f_{\text{RF}} = 990 \text{ kHz}$, and $f_{\text{LO}} = f_{\text{RF}} + f_{\text{IF}}$.

(a) $f_{\text{LO}} = 990 \text{ kHz} + 445 \text{ kHz} = 1.435 \text{ MHz.}$

(b) The image that also mixes to f_{IF} is at

$$f_{\text{image}} = f_{\text{LO}} + f_{\text{IF}} = f_{\text{RF}} + 2f_{\text{IF}} = 1.880 \text{ MHz.}$$

Centering $H_1(f)$ at f_{RF} , the two-sided 3 dB bandwidth must remain below the image offset: $B_1 < 2(f_{\text{image}} - f_{\text{RF}}) = 1.78 \text{ MHz}$ (i.e., $\pm 0.89 \text{ MHz}$ about 990 kHz keeps the image outside the passband).

(c) $w(t)$ at the image frequency sees $|H_1(f_{\text{RF}} + 2f_{\text{IF}})| = 0.1$, so its amplitude after H_1 is $3 \times 0.1 = 0.3$. The mixer output with $2 \cos(2\pi f_{\text{LO}} t)$ produces a difference tone at $|f_{\text{image}} - f_{\text{LO}}| = f_{\text{IF}}$ with amplitude 0.3 (the sum term is out of band). The ideal H_2 passes it unchanged:

$$P_{\text{int}} = \frac{0.3^2}{2} = 0.045, \quad -13.47 \text{ dBW.}$$